HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2026)

Student No.: 21 Name: _____ Marks: / 55

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Make m the subject of the formula $\frac{m}{2} - 1 = \frac{5m - n}{3}$. (3 marks) (DSE-2026 PI Q01)

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2. Simplify $\frac{(uv^2)^5}{u^7v^{-3}}$ and express your answer with positive indices. (3 marks) (DSE-2026 PI Q02)

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3. The frequency distribution table and the cumulative frequency distribution table below show the distribution of the weights of 20 apples.

Weight (grams)	Frequency
91 – 100	2
101 – 110	a
111 – 120	6
121 – 130	7
131 – 140	h

Weight less than (grams)	Cumulative frequency
100.5	2
110.5	6
120.5	12
130.5	b
140.5	k

(a) Find a and b.

(b) Find the mean of the distribution.

(5 marks) (DSE-2026 PI Q09)

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4. The straight line L passes through the point $(-10, 11)$. The slope of L is -2.

(a) Find the equation of L.

(b) The coordinates of the points H and K are $(2, 5)$ and $(-4, -7)$ respectively. Let P be a point lying on L such that P is equidistant from H and K.

(i) Find the coordinates of P.

(ii) Let Q be a point such that $HPKQ$ is a quadrilateral, where $HP = HQ = KQ$. Find the area of the quadrilateral $HPKQ$.

(7 marks) (DSE-2026 PI Q12)

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5. A solid metal sphere of radius 10 cm is melted and recast into three similar solid right circular cylinders X, Y and Z. Let r_1 cm, r_2 cm and r_3 cm be the base radii of X, Y and Z respectively. It is given that $r_1 : r_2 : r_3 = 2 : 3 : 5$.

(a) Express the volume of Y in terms of π.

(b) A craftsman finds that the height of Y is 9 cm.

(i) Find r_2.

(ii) The craftsman claims that the total surface area of Z is greater than the surface area of the sphere. Do you agree? Explain your answer.

(8 marks) (DSE-2026 PI Q13)

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6. A bag contains 3 red balls, 3 white balls and 5 black balls. If 4 balls are randomly drawn from the bag at the same time, find

(a) the probability that exactly 2 red balls are drawn;

(b) the probability that exactly 2 red balls or exactly 2 white balls are drawn.

(4 marks) (DSE-2026 PI Q15)

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7. In Figure 2, the base of the pyramid $VABCD$ is the quadrilateral $ABCD$. The face VAB of the pyramid lies on the horizontal ground. It is given that $AB = 90$ cm, $AD = 70$ cm, $CD = 100$ cm, $\angle ADC = 55°$ and $\angle BCD = 130°$.

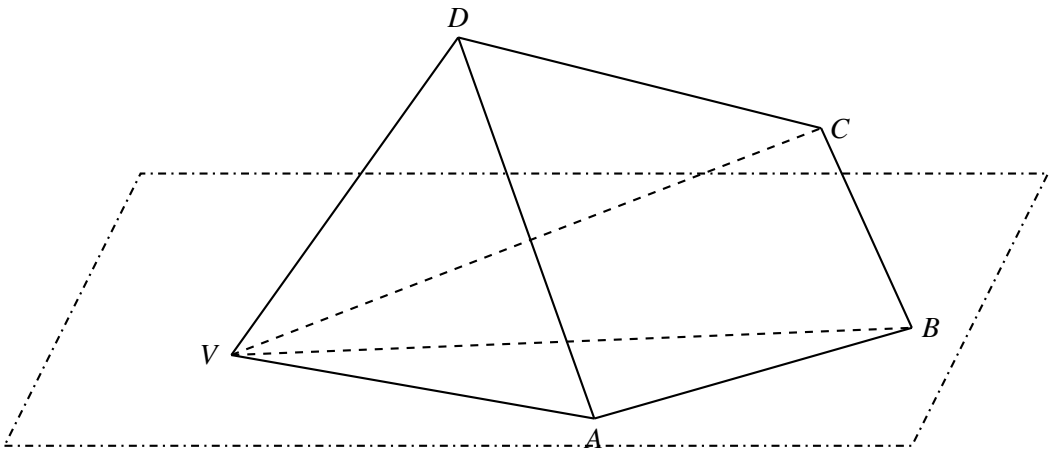


Figure 2

(a) Find

(i) the distance between A and C,

(ii) $\angle BAD$.

(b) It is given that $VD = 80$ cm. A student finds that the angle between the plane $ABCD$ and the horizontal ground is 65°. The student claims that the angle between VD and the horizontal ground is less than 50°. Is the claim correct? Explain your answer.

(6 marks) (DSE-2026 PI Q17)

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8. Let $G(n)$ be the nth term of a geometric sequence. It is given that $G(2) = 12$ and $G(5) = 768$.

(a) Find $G(1)$.

(b) Define $T(n) = \log_2 G(n)$ for all positive integers n. Find

(i) $T(n+1) - T(n)$,

(ii) the greatest value of k such that $T(1) + T(2) + T(3) + \cdots + T(k) < 10^6$.

(7 marks) (DSE-2026 PI Q18)

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9. Let $h(x) = -x^2 + 30kx + 4x - 225k^2 - 40k + 3$, where k is a positive constant. On the same rectangular coordinate system, denote the vertex of the graph of $y = h(x)$ and the vertex of the graph of $y = 14 - h(4 - x)$ by A and B respectively.

(a) Using the method of completing the square, express, in terms of k, the coordinates of A.

(b) Write down, in terms of k, the coordinates of B.

(c) The coordinates of the point C are $(2 + 48k,\ 7 - 36k)$. Let I and J be the in-centre and the circumcentre of $\triangle ABC$ respectively.

(i) Let M be the mid-point of AB. Describe the geometric relationship between CM and AB. Explain your answer.

(ii) Express, in terms of k, the distance between I and J.

(iii) The coordinates of the point D are $(6 - 22k,\ 6 - 15k)$. Is it possible that $BDIJ$ is a parallelogram? Explain your answer.

(12 marks) (DSE-2026 PI Q19)

I can finish _____ questions

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